

A LITERATURE SURVEY REPORT

ON

<WINE QUALITY CLASIFICATIONS>

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A. Y.: 2026-2027

by

Sai Lasya Grandhi 2520030538

B . Jasmitha 2520030557

Under the Esteemed guidance of

Mr. Shaik Asif

Asst.Professor, Dept. of CSE





Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

WINE QUALITY CLASSIFICATION USING LOGISTIC REGRESSION AND MULTINOMIAL LOGISTIC REGRESSION

LITERATURE SURVEY

Several studies have explored the use of statistical and machine learning techniques for predicting wine quality based on physicochemical properties. Traditional wine-quality assessment mainly depends on sensory evaluation by experts, but researchers have investigated data-driven approaches to make the process more objective and consistent. Logistic regression is one such statistical technique that can be used to study the relationship between wine characteristics and quality categories.

A relevant study, “Modeling the Preference of Wine Quality Using Logistic Regression Techniques Based on Physicochemical Properties” by Perpetual Opoku Agyemang, investigated the use of logistic regression techniques for modeling wine-quality preference. The study considered physicochemical properties of wine and examined logistic regression approaches, including ordinal and multinomial logistic regression. Multinomial logistic regression was considered when the assumptions required for the ordinal approach were not satisfied. This work demonstrates the usefulness of logistic regression for analyzing wine-quality preferences based on measurable wine characteristics.

Based on these ideas, the present project applies Binary Logistic Regression and Multinomial Logistic Regression to the UCI Red Wine Quality dataset. Binary classification is used to distinguish between Good and Not-Good wines, while multinomial classification predicts the individual quality scores. The project further uses Exploratory Data Analysis and model evaluation metrics to compare the performance of both approaches and understand the influence of physicochemical properties on wine quality.